Lecture 3

Lecture 3 builds the D1–D5–P bound state, computes the horizon area of the black hole it makes, and counts its BPS states in the brane worldvolume theory, obtaining $S = 2\pi\sqrt{Q_1 Q_5 N_p}$ twice. We then isolate what part of that computation was needed — the near-horizon AdS_3 and its central charge — and derive Cardy's formula carefully enough to see where it may be used. The remainder of the lecture develops the AdS_3/CFT_2 dictionary behind the count: the conformal manifold of the D1–D5 theory, the symmetric orbifold point and its long strings, the Seiberg–Witten singularity, and the S-dual NS–NS description, where the worldsheet theory is solvable and the boundary theory is now known exactly at $k = 1$ and in deformed form at $k \geq 2$.

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Conventions

Signature $(-,+,+)$, $\Lambda = -1/\ell^2$, $\hbar = c_{\text{light}} = 1$. The BTZ black hole is

$$\mathrm{d}s^2 = -f\,\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{f} + r^2\big(\mathrm{d}\varphi + N^\varphi \mathrm{d}t\big)^2, \qquad f = \frac{(r^2 - r_+^2)(r^2 - r_-^2)}{\ell^2 r^2}, \qquad N^\varphi = -\frac{r_+ r_-}{\ell r^2}, \tag{1}$$

with $\varphi \simeq \varphi + 2\pi$ and

$$M = \frac{r_+^2 + r_-^2}{8G\ell^2}, \quad J = \frac{r_+ r_-}{4G\ell}, \quad S = \frac{2\pi r_+}{4G}, \quad T = \frac{r_+^2 - r_-^2}{2\pi\ell^2 r_+}, \quad \Omega_H = \frac{r_-}{\ell r_+}, \quad c = \bar{c} = \frac{3\ell}{2G}. \tag{2}$$

The dual CFT_2 lives on the cylinder $\{(t,\varphi)\}$ at spatial infinity, of spatial circumference $2\pi\ell$. We measure time in units of ℓ, i.e. we use

$$x^{\pm} = \frac{t}{\ell} \pm \varphi, \tag{3}$$

so that the boundary circle has circumference 2π and all CFT quantities are dimensionless. Two sets of variables appear throughout and we fix them here.

Conformal weights. $\Delta_L \equiv L_0$ and $\Delta_R \equiv \bar{L}_0$ are the weights measured from the CFT vacuum, $\Delta = \Delta_L + \Delta_R$, spin $s = \Delta_L - \Delta_R$. The vacuum has $\Delta = 0$.

Energies. $E_L \equiv L_0 - \frac{c}{24}$, $E_R \equiv \bar{L}_0 - \frac{c}{24}$, $E = E_L + E_R = \Delta - \frac{c}{12}$. These are the energies conjugate to the modular-invariant inverse temperatures below, and the variables in which the gravitational formulas are simple. The dictionary to the bulk is

$$E_L = \frac{\ell M + J}{2} = \frac{(r_+ + r_-)^2}{16G\ell}, \qquad E_R = \frac{\ell M - J}{2} = \frac{(r_+ - r_-)^2}{16G\ell}, \tag{4}$$

[checked symbolically] so that $\ell M = E_L + E_R$ and $J = E_L - E_R$. For the non-rotating representatives:

	$E_L = E_R$	Δ	bulk
global AdS_3	$-c/24$	0	CFT vacuum, $M = -1/8G$
conical defects	$-\frac{c}{24n^2}$	$\frac{c}{12}\left(1 - \frac{1}{n^2}\right)$	$\mathbb{Z}_n$ defect, $M = -1/8Gn^2$
$M = J = 0$ geometry	0	$c/12$	Ramond ground state
Hawking–Page	$c/24$	$c/6$	$r_+ = \ell$

[checked]. The last line is derived in §2.4 (lecture 4); the first three are from lecture 2, §2.3 and [9, 10].

Inverse temperatures. The (dimensionless) potentials conjugate to E_L, E_R are

$$\beta_L = \frac{2\pi\ell}{r_+ + r_-}, \qquad \beta_R = \frac{2\pi\ell}{r_+ - r_-}, \qquad \hat{\beta} \equiv \frac{\beta_{\mathrm{phys}}}{\ell} = \frac{\beta_L + \beta_R}{2}, \qquad \hat{\beta}\,\hat{\Omega} = \frac{\beta_R - \beta_L}{2}, \tag{5}$$

with $\hat{\Omega} = \ell\Omega_H$ [checked symbolically]. In terms of the Euclidean modular parameter of the boundary torus, $\beta_L = -2\pi i\tau$, $\beta_R = 2\pi i\bar{\tau}$, so that

$$Z(\beta_L, \beta_R) = \mathrm{Tr}\, e^{-\beta_L E_L - \beta_R E_R} = \mathrm{Tr}\, q^{L_0 - \frac{c}{24}} \bar{q}^{\bar{L}_0 - \frac{c}{24}}, \qquad q = e^{2\pi i\tau} = e^{-\beta_L}, \tag{6}$$

(For $\beta_L \neq \beta_R$ the two are independent: τ and $\bar{\tau}$ are both purely imaginary but $\bar{\tau} = \tau^*$ only when $\beta_L = \beta_R$. See [39, fn. 3].) The modular S transformation $\tau \to -1/\tau$ acts as

$$\beta_L \;\to\; \beta'_L = \frac{4\pi^2}{\beta_L}, \qquad \beta_R \;\to\; \beta'_R = \frac{4\pi^2}{\beta_R}. \tag{7}$$

The self-dual point is $\beta_L = \beta_R = 2\pi$, i.e. $\beta_{\mathrm{phys}} = 2\pi\ell$, the Hawking–Page temperature of lecture 2, §1.6.

String theory. Type IIB with $16\pi G_{10} = (2\pi)^7 g_s^2 \alpha'^4$, i.e. $G_{10} = 8\pi^6 g_s^2 \alpha'^4$. The volume of a compact T^4 is written $V = (2\pi)^4 v\, \alpha'^2$ with v dimensionless, so that the six-dimensional string coupling is $g_6^2 = g_s^2/v$.

1 The D1–D5–P system and the Strominger–Vafa entropy

1.1 The bound state, the black hole, and the microscopic count

The configuration

Compactify type IIB string theory on

$$\mathbb{R}^{1,4} \;\times\; S^1_y \;\times\; M_4, \qquad M_4 = T^4 \text{ or } K3, \tag{8}$$

with S^1_y of radius R and M_4 of volume V, and wrap

$$\begin{array}{ll} Q_5 \text{ D5-branes} & \text{on } S^1_y \times M_4, \\ Q_1 \text{ D1-branes} & \text{on } S^1_y, \\ N_p \text{ units of momentum} & \text{along } y. \end{array} \tag{9}$$

Each ingredient separately preserves 16 of the 32 supercharges; the D1–D5 pair preserves 8; adding the momentum wave leaves 4. The configuration is $\frac{1}{8}$-BPS, which is what makes the computation possible: the BPS degeneracy is computed by an index, and an index does not change as g_s is varied continuously — provided no states run off to infinity, a caveat that returns in §2.6.

From the five-dimensional point of view — after reducing on $S^1_y \times M_4$ — this is a static, extremal black hole with three charges. From the six-dimensional point of view it is a black *string* wound on S^1_y; that second picture is the one relevant to AdS_3 and we return to it in §1.2.

The supergravity solution and its horizon

The ten-dimensional string-frame solution is built from three harmonic functions on the transverse $\mathbb{R}^4$,

$$H_1 = 1 + \frac{r_1^2}{r^2}, \qquad H_5 = 1 + \frac{r_5^2}{r^2}, \qquad H_p = 1 + \frac{r_p^2}{r^2}, \tag{10}$$

with

$$\boxed{\,r_1^2 = \frac{g_s \alpha' Q_1}{v}, \qquad r_5^2 = g_s \alpha' Q_5, \qquad r_p^2 = \frac{g_s^2 \alpha'^2 N_p}{v\, R^2}\,} \tag{11}$$

in the conventions fixed above [2, 6, 8]. These are not independent normalisations. They follow from the standard Dp-brane harmonic function

$$r_{\mathrm{D}p}^{7-p} = (2\sqrt{\pi})^{5-p}\, \Gamma\!\Big(\tfrac{7-p}{2}\Big)\, g_s N\, (\alpha')^{(7-p)/2}, \tag{12}$$

which for $p = 5$ gives $r_5^2 = g_s \alpha' Q_5$ directly, the D5 being transverse only to $\mathbb{R}^4$. For the D1 and the momentum wave one must first smear over M_4, using

$$\int_{\mathbb{R}^4} \frac{\mathrm{d}^4 y}{(r^2 + y^2)^3} = \frac{\pi^2}{2 r^2} \qquad \Longrightarrow \qquad \frac{c_{\mathrm{D1}}}{|x|^6} \;\longrightarrow\; \frac{\pi^2 c_{\mathrm{D1}}}{2V\, r^2}, \tag{13}$$

[checked] and with $c_{\mathrm{D1}} = r_{\mathrm{D1}}^6 = 32\pi^2 g_s Q_1 \alpha'^3$ this gives $r_1^2 = 16\pi^4 g_s Q_1 \alpha'^3 / V = g_s \alpha' Q_1 / v$ [checked]. The factor $1/v$ that distinguishes r_1, r_p from r_5 is exactly this smearing. The momentum wave carries no independent tension, and r_p^2 is fixed instead by requiring that it contribute N_p/R to the mass; we verify this below.

The five-dimensional Einstein-frame metric obtained by reducing on $S^1_y \times M_4$ is

$$\mathrm{d}s_5^2 = -\big(H_1 H_5 H_p\big)^{-2/3} \mathrm{d}t^2 + \big(H_1 H_5 H_p\big)^{1/3} \big(\mathrm{d}r^2 + r^2 \mathrm{d}\Omega_3^2\big). \tag{14}$$

The horizon is at $r = 0$. Although each metric function degenerates there, the S^3 keeps a finite radius: as $r \to 0$,

$$(H_1 H_5 H_p)^{1/3} r^2 \;\longrightarrow\; \left(\frac{r_1^2 r_5^2 r_p^2}{r^6}\right)^{1/3} r^2 = (r_1 r_5 r_p)^{2/3}, \tag{15}$$

so the horizon is a round S^3 of radius $(r_1 r_5 r_p)^{1/3}$ and area

$$A_3 = 2\pi^2\, r_1 r_5 r_p = 2\pi^2\, \frac{g_s^2 \alpha'^2}{v\, R} \sqrt{Q_1 Q_5 N_p} \tag{16}$$

[checked]. The five-dimensional Newton constant is

$$G_5 = \frac{G_{10}}{V \cdot 2\pi R} = \frac{8\pi^6 g_s^2 \alpha'^4}{(2\pi)^4 v \alpha'^2 \cdot 2\pi R} = \frac{\pi g_s^2 \alpha'^2}{4 v R} \tag{17}$$

[checked], and the entropy is

$$\boxed{S_{\rm BH} = \frac{A_3}{4G_5} = \frac{2\pi^2}{4} \cdot \frac{g_s^2 \alpha'^2}{vR} \sqrt{Q_1 Q_5 N_p} \cdot \frac{4vR}{\pi g_s^2 \alpha'^2} = 2\pi \sqrt{Q_1 Q_5 N_p}} \tag{18}$$

[checked]. Every parameter of the compactification has cancelled: g_s, v, R and α' all drop out, and the entropy is a function of the three integer charges alone. Nothing forced this — it is a property of the three-charge extremal solution — and it is what allows a horizon area computed at $g_s^2 Q_i \gg 1$ to be compared with a degeneracy computed at $g_s \to 0$.

Exercise 1. Repeat the computation for the two-charge system, $N_p = 0$. Show that $A_3 = 0$: the two-charge extremal solution has a null singularity rather than a regular horizon. Three charges are the minimum for a macroscopic horizon in five dimensions, and the missing entropy of the two-charge system is the starting point of the fuzzball programme [7].

The worldvolume theory

Now go to $g_s \to 0$ at fixed Q_1, Q_5, N_p, where the branes are flat and the geometry (14) is negligible. The massless open-string degrees of freedom are the 1–1, 5–5 and 1–5 strings. Reducing on M_4 and keeping the directions along $\mathbb{R}_t \times S^1_y$, one obtains a $1+1$ dimensional gauge theory with $\mathcal{N} = (4,4)$ supersymmetry: a $U(Q_1) \times U(Q_5)$ vector multiplet, adjoint hypermultiplets from the 1–1 and 5–5 strings, and bifundamental hypermultiplets from the 1–5 strings.

The theory has two branches.

- The **Coulomb branch**: the 1–5 hypermultiplets vanish and the adjoint scalars are switched on, i.e. the D1-branes sit at separated points of the transverse $\mathbb{R}^4$, away from the D5-branes.
- The **Higgs branch**: the 1–5 hypermultiplets are switched on, i.e. the D1-branes are dissolved into the D5-branes. A D1 inside a D5 is a finite-size instanton of the D5 gauge field [5, 4, 3], so the Higgs branch is

$$\mathcal{M}_{Q_1,Q_5} = \big\{\text{moduli space of } Q_1 \text{ instantons of } U(Q_5) \text{ on } M_4\big\}, \tag{19}$$

  a hyperkähler manifold. For compact M_4 its real dimension is $4(Q_1 Q_5 + 1)$; the ADHM count on $\mathbb{R}^4$ would give $4Q_1 Q_5$, and the extra quaternionic direction is a compactness effect.

The bound state lives on the Higgs branch, and in the infrared the gauge theory flows to the $\mathcal{N} = (4,4)$ sigma model with target (19). Its central charge is fixed by the target dimension — each real boson contributes 1 and each Majorana–Weyl fermion $\frac{1}{2}$, and $(4,4)$ supersymmetry pairs them:

$$c = \tfrac{3}{2}\,\dim_{\mathbb{R}} \mathcal{M}_{Q_1,Q_5} = 6(Q_1Q_5 + 1). \tag{20}$$

For $M_4 = T^4$ four of these directions — the centre of mass of the bound state in T^4, together with its dual — decouple as a free $c = 6$ theory, and the interacting part has

$$\boxed{c = 6\,Q_1Q_5} \qquad (M_4 = T^4), \tag{21}$$

while for $M_4 = K3$ there is no such decoupling and $c = 6(Q_1Q_5+1)$. For $Q_1Q_5 \gg 1$ the difference is invisible and we write $c = 6N$ with $N = Q_1Q_5$ throughout. [checked]

The count

From the point of view of the $1+1$ dimensional theory on $\mathbb{R}_t \times S^1_y$, momentum along y is the generator of translations along the spatial circle:

$$N_p = L_0 - \bar{L}_0 \qquad \text{and, for a BPS state,} \qquad \bar{L}_0 = \frac{c}{24}, \quad L_0 = \frac{c}{24} + N_p, \tag{22}$$

i.e. $E_L = N_p$, $E_R = 0$. (The right-movers sit in their Ramond ground state; that is what supersymmetry means here.) What we need is therefore the degeneracy $\rho(E_L = N_p,\, E_R = 0)$ of a CFT with $c = 6Q_1Q_5$ at left-moving level N_p. For $N_p \gg c$ this is Cardy's formula, derived in §1.2 below:

$$S_{\text{micro}} = 2\pi\sqrt{\frac{c\,N_p}{6}} = 2\pi\sqrt{\frac{6Q_1Q_5\,N_p}{6}} = 2\pi\sqrt{Q_1Q_5N_p}, \tag{23}$$

in agreement with the supergravity answer (18) [1]. [checked]

What is being compared. The two computations are done at opposite ends of the parameter range. Equation (18) needs the curvature at the horizon to be small in string units, which by (26) below means $\ell^2/\alpha' = g_6\sqrt{Q_1Q_5} \gg 1$; (23) needs the open-string description to be weakly coupled, $g_sQ_i \ll 1$. There is no overlap. The bridge is supersymmetry: the quantity that is actually protected is a right-moving index — concretely the elliptic genus (49) — which is invariant under continuous variations of the couplings. Strictly, then, what one may transport across moduli space is the index, not the absolute degeneracy; the two have the same exponential growth at large charge because the cancellations in $\mathrm{Tr}(-1)^F$ are subleading, which is why counting all left-moving states at level N_p is legitimate here.
What is *not* protected is the non-BPS spectrum, the free energy at finite temperature, or anything about a non-extremal black hole. The Strominger–Vafa argument as it stands says nothing about those; supplying the missing statements is the content of lecture 4.

Check: the mass. Expanding (14) at large r, $g_{tt} = -(H_1H_5H_p)^{-2/3} \to -\big(1 - \frac{2}{3}\sum_i r_i^2/r^2\big)$, and comparing with the five-dimensional asymptotic form $g_{tt} = -\big(1 - 8G_5M/3\pi r^2\big)$ gives $M = \pi\sum_i r_i^2/4G_5$, i.e.

$$M_{\text{ADM}} = \frac{R}{g_s\alpha'}\big(Q_1 + v\,Q_5\big) + \frac{N_p}{R} \tag{24}$$

[checked symbolically] with the radii (11). The first two terms are the tensions $T_{\text{D1}} = 1/(2\pi\alpha' g_s)$ and $T_{\text{D5}} = 1/((2\pi)^5\alpha'^3 g_s)$ times the wrapped volumes $2\pi R$ and $2\pi R\,V$ [checked], and the third is the momentum N_p/R — which is what fixes r_p^2. The three charges contribute additively with no binding energy, and this is why the degeneracy factorises into left-movers at level N_p times a right-moving ground state.

1.2 The near-horizon geometry, Cardy's formula, and Strominger's derivation

The count of §1.1 used a great deal of string theory: a particular brane bound state, a particular instanton moduli space, a particular supersymmetry algebra. We now isolate the part that was actually necessary. The answer, due to Strominger [9], is that only two numbers entered — the central charge and the level — and that the central charge can be read off from the near-horizon geometry using Brown–Henneaux, with no reference to branes.

The near-horizon limit

Return to the six-dimensional picture, before compactifying y. Drop the 1's in (10) — i.e. take $r \ll r_1, r_5$ — and set $N_p = 0$ for the moment. The six-dimensional string-frame metric is

$$\mathrm{d}s_6^2 = (H_1H_5)^{-1/2}\big(-\mathrm{d}t^2 + \mathrm{d}y^2\big) + (H_1H_5)^{1/2}\big(\mathrm{d}r^2 + r^2\mathrm{d}\Omega_3^2\big) \;\longrightarrow\; \frac{r^2}{\ell^2}\big(-\mathrm{d}t^2 + \mathrm{d}y^2\big) + \ell^2\frac{\mathrm{d}r^2}{r^2} + \ell^2\mathrm{d}\Omega_3^2, \tag{25}$$

[checked]: Poincaré AdS_3 and S^3 with the same radius,

$$\ell^2 = r_1 r_5 = \sqrt{\frac{g_s\alpha' Q_1}{v}\cdot g_s\alpha' Q_5} = g_6\,\alpha'\sqrt{Q_1Q_5}, \qquad g_6 = \frac{g_s}{\sqrt{v}}, \tag{26}$$

[checked], times M_4. Two features of the near-horizon region:

- the dilaton is constant, $e^{2\phi} \to g_s^2 r_1^2/r_5^2$;
- the M_4 metric is warped by $(H_1/H_5)^{1/2}$, so its volume flows to the attractor value $V_{\rm nh} = V\,r_1^2/r_5^2$, i.e. $v_{\rm nh} = Q_1/Q_5$ [checked symbolically] — fixed by the charges, independently of where one started.

The two shifts compensate in the six-dimensional string coupling,

$$g_6^2\big|_{\text{near hor.}} = \frac{(2\pi)^4 e^{2\phi}\alpha'^2}{V_{\rm nh}} = \frac{g_s^2}{v} = g_6^2 \tag{27}$$

[checked symbolically]: unlike v, the coupling g_6 is *not* fixed by the attractor, and remains a modulus of the near-horizon solution. The background is weakly coupled whenever g_6 is small.

The central charge from the bulk

Reduce on the S^3, of volume $2\pi^2\ell^3$, and on M_4:

$$G_6 = \frac{G_{10}}{V} = \frac{8\pi^6 g_s^2\alpha'^4}{(2\pi)^4 v\alpha'^2} = \frac{\pi^2 g_6^2\alpha'^2}{2}, \qquad G_3 = \frac{G_6}{2\pi^2\ell^3}, \tag{28}$$

[checked]. (By the previous paragraph the near-horizon shifts of $e^{-2\phi}$ and of the M_4 volume cancel, so (28) may be used with the asymptotic values.) Brown–Henneaux then gives a formula we use twice:

$$\boxed{c = \frac{3\ell}{2G_3} = \frac{3\pi^2\ell^4}{G_6} = \frac{6\,\ell^4}{g_6^2\,\alpha'^2}} \tag{29}$$

[checked]. With (26),

$$c = \frac{6\,g_6^2\alpha'^2 Q_1Q_5}{g_6^2\alpha'^2} = 6\,Q_1Q_5, \tag{30}$$

matching (21), which came from counting the dimension of an instanton moduli space. The same result in Strominger's variables [9, §6] reads $\ell = 2\pi\alpha'\sqrt{g_s}(Q_1Q_5)^{1/4}/V^{1/4}$ and $G_3^{-1} = 2V^{1/4}(Q_1Q_5)^{3/4}/(\pi\alpha'\sqrt{g_s})$, whose product gives $\frac{3}{2}\cdot 4\,Q_1Q_5$ [checked].

Note what (30) did *not* use: no supersymmetry, no instantons, no D-branes. It used the near-horizon metric and Brown–Henneaux. Any black hole with a locally-AdS_3 near-horizon region will do.

Turning the momentum back on

With $N_p \neq 0$ the near-horizon geometry is still locally AdS_3 — it must be, since every solution of $3d$ vacuum gravity is (lecture 1, §1.1) — but the global identification changes. Compactifying y means quotienting the near-horizon AdS_3 by a discrete isometry, and the resulting quotients are the family constructed in lecture 1, §2.3:

- $N_p = 0$: the identification $y \simeq y + 2\pi R$ acts on Poincaré AdS_3 by a *parabolic* element, and the quotient is the $M = J = 0$ geometry, $r_+ = r_- = 0$.
- $N_p > 0$ with the right-movers in their ground state (the BPS case): $E_L = N_p$, $E_R = 0$, hence $r_+ = r_-$ by (4). One gets an *extremal* BTZ black hole, $\ell M = J$.
- A generic, non-extremal BTZ needs $E_L \neq 0$ *and* $E_R \neq 0$, i.e. momentum in both chiralities. The corresponding brane configuration carries N_L left-moving and N_R right-moving quanta with net momentum $N_p = N_L - N_R$, and is not BPS.

The three-charge BPS system therefore accounts only for the extremal corner of the BTZ family. Non-extremal black holes are described by the same CFT, but by non-BPS states in it, whose degeneracy is not an index and is not protected as g_s is varied. For those one needs the arguments of lecture 4.

Cardy's formula

We have used Cardy's formula twice already; we now derive it, and determine when it may be used.

Let $\mathcal{T}$ be a unitary CFT_2 with a discrete spectrum, central charges $c = \bar{c}$, and a normalisable $SL(2,\mathbb{C})$-invariant vacuum. Put it on a torus and write

$$Z(\beta_L,\beta_R) = \sum_{\Delta_L,\Delta_R} \rho(\Delta_L,\Delta_R)\, e^{-\beta_L\left(\Delta_L - \frac{c}{24}\right) - \beta_R\left(\Delta_R - \frac{c}{24}\right)} = \sum \rho\, e^{-\beta_L E_L - \beta_R E_R}. \tag{31}$$

The torus with modular parameter τ and the torus with $-1/\tau$ are the same torus, described by exchanging the roles of "space" and "Euclidean time". Hence

$$Z(\beta_L,\beta_R) = Z\Big(\frac{4\pi^2}{\beta_L}, \frac{4\pi^2}{\beta_R}\Big) \equiv Z(\beta'_L,\beta'_R). \tag{32}$$

(That the real-β version follows from the complex one is a short holomorphy argument, given in [39, fn. 3].)

Now take $\beta_L, \beta_R \to 0$. The left-hand side is a sum over all states at very high temperature and is intractable. The right-hand side is at very low temperature, $\beta'_{L,R} \to \infty$, where the sum is dominated by the state of lowest E: the vacuum, at $E_L = E_R = -c/24$. Therefore

$$Z(\beta_L,\beta_R) \;\simeq\; \exp\Big[\frac{c}{24}\big(\beta'_L + \beta'_R\big)\Big] = \exp\Big[\frac{\pi^2 c}{6}\Big(\frac{1}{\beta_L} + \frac{1}{\beta_R}\Big)\Big]. \tag{33}$$

Invert the Laplace transform,

$$\rho(E_L,E_R)=\frac{1}{(2\pi i)^2}\int \mathrm{d}\beta_L\,\mathrm{d}\beta_R\; e^{\beta_L E_L+\beta_R E_R}\,Z(\beta_L,\beta_R), \tag{34}$$

and evaluate by saddle point. The two chiralities decouple; for the left one the exponent is

$$f(\beta_L)=\beta_L E_L+\frac{\pi^2 c}{6\beta_L},\qquad f'(\beta_L)=E_L-\frac{\pi^2 c}{6\beta_L^2}=0\quad\Longrightarrow\quad \beta_L^*=\pi\sqrt{\frac{c}{6E_L}}, \tag{35}$$

and the two terms in f are equal at the saddle, each contributing $\pi\sqrt{cE_L/6}$. Hence

$$\boxed{S_{\text{Cardy}}(E_L,E_R)=\log\rho=2\pi\sqrt{\frac{c\,E_L}{6}}+2\pi\sqrt{\frac{\bar{c}\,E_R}{6}}} \tag{36}$$

[checked], the Gaussian fluctuation about the saddle contributing only $-\frac{3}{4}\log E_L+\dots$.

Where the formula may be used. Two approximations were made, and they fail for different reasons.

(i) The saddle must lie in the high-temperature regime. Passing to the dual channel is only useful if β_L^* is small, i.e. if $4\pi^2/\beta_L^*$ is large, so that the dual channel really is cold. The dividing line is the self-dual point $\beta_L^*=2\pi$:

$$\beta_L^*<2\pi\quad\Longleftrightarrow\quad \pi\sqrt{\frac{c}{6E_L}}<2\pi\quad\Longleftrightarrow\quad \boxed{E_L>\frac{c}{24}} \tag{37}$$

[checked], and similarly for E_R. Below $c/24$ the saddle sits on the wrong side of the self-dual temperature and the derivation does not apply.

(ii) The dual channel must be vacuum dominated. The corrections to (33) are

$$Z(\beta_L',\beta_R')=e^{\frac{c}{24}(\beta_L'+\beta_R')}\Big[1+\sum_{\Delta>0}\rho(\Delta_L,\Delta_R)\,e^{-\beta_L'\Delta_L-\beta_R'\Delta_R}\Big], \tag{38}$$

where the sum runs over non-vacuum states and we used $E_{L,R}+\frac{c}{24}=\Delta_{L,R}$. Whether the bracket is close to 1 depends on how many light states there are. *At fixed* c, taking $E_L\to\infty$ sends $\beta_L'\to\infty$ and the bracket to 1: this is the classical Cardy limit, and (36) is then a theorem. But in the holographic limit we take $c\to\infty$ with E_L/c fixed, in which case β_L' is of order 2π and the bracket is

$$1+\sum_{\Delta>0}\rho(\Delta)\,e^{-2\pi\Delta}+\dots, \tag{39}$$

which is close to 1 only if the density of light states satisfies $\rho(\Delta)\lesssim e^{2\pi\Delta}$. That condition is not automatic and is not implied by anything said so far. It is the sparseness assumption of Hartman–Keller–Stoica, and §2 (lecture 4) shows that it is both necessary and sufficient.

The conservative statement. Unwilling to assume anything about the light spectrum, one falls back on the condition Strominger states [9, eq. (5.1)]:

$$E_L,E_R\;\gg\;c. \tag{40}$$

Then $\beta_L'=4\pi\sqrt{6E_L/c}\to\infty$, and the sum in (38) is suppressed by $e^{-\beta'\Delta}$ whatever the light spectrum: any fixed exponential growth rate $\rho\sim e^{a\Delta}$ is eventually beaten once $\beta'>a$ [checked]. This is safe, but as §1.3 explains it is far more restrictive than necessary.

The entropy of BTZ from Cardy

Combine (36) with the Brown–Henneaux central charge and the identification (4), all of which are statements about $3d$ gravity alone:

$$S_{\text{Cardy}} = 2\pi\sqrt{\frac{c}{6}\cdot\frac{(r_+ + r_-)^2}{16G\ell}} + 2\pi\sqrt{\frac{c}{6}\cdot\frac{(r_+ - r_-)^2}{16G\ell}} = \frac{\pi(r_+ + r_-)}{4G} + \frac{\pi(r_+ - r_-)}{4G} = \frac{\pi r_+}{2G} = \frac{A}{4G} \tag{41}$$

[checked symbolically], for *every* BTZ black hole; equivalently

$$S = \pi\sqrt{\frac{\ell(\ell M + J)}{2G}} + \pi\sqrt{\frac{\ell(\ell M - J)}{2G}}, \tag{42}$$

which is [9, eq. (5.3)]. Applied to the BPS configuration $E_L = N_p$, $E_R = 0$ with $c = 6Q_1Q_5$ this is (23), so Strominger–Vafa is a special case. But (41) did not use M_4, the instanton moduli space, or supersymmetry; it used only that the near-horizon geometry is AdS_3 and that $c = 3\ell/2G$. Any consistent quantum theory of gravity with these asymptotics — with or without matter, string-derived or not — gives the same answer.

The inputs of the argument. Four assumptions have been made, and each is revisited later in the course.

(1) A CFT *exists.* Brown–Henneaux delivers an *algebra* and a central charge. Cardy's formula additionally requires a unitary, modular-invariant theory with a discrete spectrum. The Liouville example of lecture 2, §5.2 shows this is a real assumption: Liouville has $c = 3\ell/2G$, but a continuous spectrum bounded below by $\Delta_{\min} = (c-1)/24$, so that its effective central charge is $c_{\text{eff}} = c - 24\Delta_{\min} = 1$; (40) (lecture 4) with $c \to c_{\text{eff}}$ then gives $S \sim 2\pi\sqrt{E/3}$ [checked], nowhere near $A/4G$. A central charge alone does not determine a density of states.

(2) The ground state is global AdS_3. The entire exponent in (33) came from the vacuum sitting at $E = -c/24$ in each chirality. What controls the growth is therefore $c_{\text{eff}} = c - 24\Delta_{\min}$, with $\Delta_{\min}$ the lowest eigenvalue of L_0: getting $A/4G$ requires $c_{\text{eff}} = c$, i.e. a normalisable state at $\Delta_{\min} = 0$, which is global AdS_3. Had the lightest state instead been the $M = 0$ geometry, at $\Delta = c/12$, the dual channel would have been dominated by $e^{-\beta'(c/12 - c/12)} = 1$ rather than by $e^{\beta' c/12}$, and there would be no Cardy growth at all. In the D1–D5 realisation the required statement is that the Neveu–Schwarz vacuum, not the Ramond ground state, is the true ground state [9, 10]; in pure gravity it is an assumption about the spectrum of conical defects.

(3) The regime of validity. Discussed above and in §1.3.

(4) The states are counted at infinity. The states in (36) are excitations of the CFT on the cylinder at $r = \infty$. Nothing localises them at the horizon and nothing distinguishes one microstate from another. This is both the strength of the derivation — it is insensitive to the interior, hence universal — and its limitation [11, 12].

1.3 The regime-of-validity problem

Condition (40), $E_{L,R} \gg c$, translates in the bulk into

$$\frac{(r_+ \pm r_-)^2}{16G\ell} \gg \frac{3\ell}{2G} \qquad \Longleftrightarrow \qquad r_+ \pm r_- \gg \ell, \tag{43}$$

[checked]: black holes much larger than the AdS scale. That is not the regime in which the semiclassical description holds. A BTZ black hole is a good classical solution as soon as $S = \pi r_+/2G \gg 1$, i.e. as soon as $c\,(r_+/\ell) \gg 1$, which at $r_+ \sim \ell$ requires only $c \gg 1$. So (40) discards, for no good reason:

- every black hole near the Hawking–Page transition, $r_+ = \ell$;
- every near-extremal black hole, $r_+ \to r_-$, where $E_R \to 0$;
- in the Strominger–Vafa setting, everything except $N_p \gg Q_1 Q_5$ — which is not where the five-dimensional black hole (14) is semiclassical. That requires $g_6\sqrt{Q_1 Q_5} \gg 1$ and $S_{\rm BH} = 2\pi\sqrt{Q_1 Q_5 N_p} \gg 1$, neither of which is a condition of the form (40).

There are two ways out, and we use both.

Use the structure of the dual theory. If the boundary theory is a symmetric orbifold $\mathrm{Sym}^N(\mathcal{C})$ — as it is at the orbifold point of the D1–D5 conformal manifold — then the maximally twisted sector contains a single "long string" of length N, whose modes have spacing $1/N$. A state at level E_L of the full theory sits at effective level NE_L in a CFT of central charge $c_1 = c/N$, so Cardy applied to the long string requires

$$NE_L \gg c_1 = \frac{c}{N} \qquad \Longleftrightarrow \qquad E_L \gg \frac{c}{N^2}, \tag{44}$$

[checked] rather than $E_L \gg c$: the window has widened by N^2. This is the long-string mechanism of Maldacena–Susskind [13], made precise in §2.3 (lecture 4).

Assume nothing about the dual theory. Assume only large c, modular invariance, unitarity, and a light spectrum sparse in the sense of (38). Then (36) holds for all $\Delta \geq c/6$. This is [39], developed in §2 (lecture 4); it applies to $3d$ gravity whether or not the bulk is a string compactification.

2 The D1–D5 correspondence in detail

2.1 Statement of the duality

Take the decoupling limit of the D1–D5 system: $\alpha' \to 0$ with $U \equiv r/\alpha'$, g_6, v and $R/\sqrt{\alpha'}$ held fixed. The asymptotically flat region decouples and only the near-horizon throat survives. Maldacena's conjecture [14] here reads

$$\boxed{\begin{array}{c}\text{IIB string theory on } \mathrm{AdS}_3 \times S^3 \times M_4 \\ \text{with } Q_5 \text{ units of RR 3-form flux on } S^3\end{array} \quad \Longleftrightarrow \quad \begin{array}{c}\text{2d } \mathcal{N} = (4,4) \text{ SCFT} \\ \text{with } c = \bar{c} = 6Q_1Q_5\end{array}} \tag{45}$$

The symmetry match is the first check. The bulk isometries are

$$\mathrm{SO}(2,2) \times \mathrm{SO}(4)_{S^3} = \big[\mathrm{SL}(2,\mathbb{R})_L \times \mathrm{SL}(2,\mathbb{R})_R\big] \times \big[\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R\big], \tag{46}$$

exactly the bosonic part of the *global* small $\mathcal{N} = (4,4)$ superconformal algebra: two $\mathfrak{sl}(2)$'s and two $\mathfrak{su}(2)$ R-symmetries. On the boundary the $\mathfrak{sl}(2)$'s are enhanced to Virasoro (Brown–Henneaux) and the $\mathfrak{su}(2)$'s to affine $\mathfrak{su}(2)$ at some level k'; the bulk counterpart of the second enhancement is the asymptotic symmetry analysis of the S^3 gauge fields [17, 19]. The small $\mathcal{N} = 4$ algebra ties the two together,

$$c = 6k', \tag{47}$$

so $k' = Q_1Q_5 = N$: the R-symmetry level is not an independent parameter. This is used in §2.5.

2.2 The conformal manifold

Both sides of (45) come in families. On the bulk side, type IIB on M_4 gives a six-dimensional supergravity whose scalars parametrise $\mathcal{O}(5,n)/(\mathcal{O}(5)\times\mathcal{O}(n))$, of dimension $5n$, with $n=5$ for T^4 and $n=21$ for $K3$: the metric and B-field on M_4, the RR potentials on M_4, and g_6. Switching on Q_5 units of three-form flux singles out a direction in the $\mathcal{O}(5)$ factor, and the corresponding n scalars are fixed at the horizon by the attractor mechanism — $v_{\rm nh}=Q_1/Q_5$ of §1.2 is one of them. The remaining $4n$ are genuine moduli of the AdS_3 vacuum. On the boundary side, an exactly marginal operator — a $(1,1)$ primary that remains $(1,1)$ once switched on — may be added to the action without spoiling conformal invariance, and the space of couplings so generated is the *conformal manifold* $\mathcal{M}$: the space of inequivalent CFT's reached from a given one by such deformations. Each surviving bulk modulus is a coordinate on $\mathcal{M}$, and the duality is a statement about the whole family. Locally [20, 21, 22]

$$\mathcal{M} \;\simeq\; \mathcal{O}(4,n;\mathbb{Z})\backslash\frac{\mathcal{O}(4,n)}{\mathcal{O}(4)\times\mathcal{O}(n)}, \qquad n=\begin{cases}5, & M_4=T^4, \quad \dim_{\mathbb{R}}\mathcal{M}=20,\\ 21, & M_4=K3, \quad \dim_{\mathbb{R}}\mathcal{M}=84.\end{cases} \tag{48}$$

The dimensions $4n=20$ and 84 are the $5n=25$ and 105 six-dimensional scalars minus the n fixed by the attractor [checked]. The structure is Narain-like because the moduli organise into a Grassmannian of 4-planes; the arithmetic quotient is the duality group identifying physically equivalent points.

Three consequences run through everything below.

(a) Only protected quantities can be compared across $\mathcal{M}$. The spectrum changes as one moves on $\mathcal{M}$: dimensions of non-BPS operators are functions of the moduli. What does not change is the elliptic genus

$$\chi(\tau,z)=\mathrm{Tr}_{RR}\Big[(-1)^F q^{L_0-\frac{c}{24}}\bar q^{\bar L_0-\frac{c}{24}}\,y^{2J_0^3}\Big], \qquad q=e^{2\pi i\tau},\ y=e^{2\pi i z}. \tag{49}$$

Only right-moving ground states contribute, the $\bar q$ dependence cancels, and the result is a weak Jacobi form of weight 0 and index $c/6$, constant on $\mathcal{M}$ — with the caveat, spelled out in §2.6, that this can fail on loci where states escape to infinity in field space. This is the technical content of the "BPS states are protected" statement used in §1.1. For $\mathrm{Sym}^N(K3)$ the generating function of elliptic genera was computed in [23] and compared with the supergravity spectrum by de Boer [18]: the two agree in the range of low levels where supergravity is expected to be valid, and part company above it, where the orbifold has far more states.

(b) There is a distinguished point: the orbifold point. For one choice of moduli the interacting sigma model on $\mathcal{M}_{Q_1,Q_5}$ degenerates to a free orbifold,

$$\boxed{\mathcal{T}_{\rm orb}=\mathrm{Sym}^N(M_4)\equiv\Big(\text{sigma model with target } (M_4)^N/S_N\Big), \qquad N=Q_1Q_5, \qquad c=6N.} \tag{50}$$

Geometrically this is the statement that $\mathcal{M}_{Q_1,Q_5}$ is deformation equivalent to a resolution of $(M_4)^N/S_N$ [3, 21]. For $M_4=T^4$ the resulting theory is free — $4N$ bosons and $4N$ fermions per chirality, subject to the S_N projection and twisted sectors — and for $M_4=K3$ it is a free orbifold of the (non-free but solvable) $K3$ sigma model.

(c) Parts of $\mathcal{M}$ are singular. This is §2.6.

2.3 Where the bulk is stringy

Point (b) is why anything can be computed, and also why one must be careful about what the computation means: at the orbifold point the bulk is a *tensionless* string, not Einstein gravity. Three independent ways to see this.

Higher-spin currents. A free CFT has infinitely many conserved currents. In $\mathrm{Sym}^N(T^4)$ the single-copy fields $\partial X^i, \psi^i$ combine into S_N-invariant holomorphic currents at every integer spin $s \geq 2$ (and, from the fermions, at half-integer spin as well),

$$J_s \;\sim\; \sum_{a=1}^{N} \big(\partial X^i \partial^{s-1} X^j\big)_a + \ldots, \qquad \bar\partial J_s = 0, \qquad \Delta = s. \tag{51}$$

A conserved spin-s current of dimension $\Delta = s$ is dual to a *massless* spin-s gauge field in the bulk. A theory of gravity with massless fields of arbitrarily high spin is not Einstein gravity; it is the tensionless limit of a string, in which the entire leading Regge trajectory becomes massless. Deforming away from the orbifold point gives the currents with $s > 2$ anomalous dimensions, $\Delta_s = s + \gamma_s(\lambda)$ with $\gamma_s \neq 0$: the higher-spin fields eat Goldstone modes and become massive, and the string acquires tension [28, 29].

The string length. The free symmetric orbifold $\mathrm{Sym}^N(T^4)$ is the boundary dual of the pure NS–NS background at $k = 1$ (§3.3), where

$$\frac{\ell^2}{\alpha'} = k = 1. \tag{52}$$

The AdS radius equals the string length, so there is no regime in which the α' expansion converges.

Thermodynamics. A tower of string states at the AdS scale produces a density of light states growing exponentially in Δ. We show in §2.3 (lecture 4) that $\mathrm{Sym}^N(\mathcal{C})$ has exactly $\rho(\Delta) \sim e^{2\pi\Delta}$ for light Δ: Hagedorn growth, with the Hagedorn temperature sitting precisely at the Hawking–Page temperature. A bulk consisting of Einstein gravity plus finitely many light fields would instead give $\log \rho \sim \Delta^{2/3}$ (§1.3 (lecture 4)).

None of this invalidates the entropy match of §1.1: the elliptic genus is protected, so the BPS count is the same at the orbifold point and in the supergravity regime. What it does mean is that the orbifold point is not itself a semiclassical gravity dual, and that statements about unprotected quantities computed there need not survive the trip across $\mathcal{M}$.

The deformation. Moving off the orbifold point requires a twisted-sector modulus. Counting: $\dim_{\mathbb{R}} \mathcal{M} = 20$ for T^4, of which 16 are the Narain moduli $\mathcal{O}(4,4)/\mathcal{O}(4)^2$ of the seed; the remaining four live in the $\mathbb{Z}_2$-twisted sector and are the blow-up modes of the orbifold singularity. (For $K3$: $84 = 80 + 4$, with the same four.) [checked]They are superconformal descendants of the two-cycle twist field,

$$\mathcal{O}_2 \;\sim\; G_{-1/2}\bar G_{-1/2}\,\sigma_2, \qquad (h,\bar h) = (1,1), \qquad \delta S = \lambda \int \mathrm{d}^2 z\; \mathcal{O}_2(z,\bar z), \tag{53}$$

and we write $\mathcal{O}_2$ for the combination that preserves the $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ R-symmetry [21, 20, 27]. This is the deformation that lifts the higher-spin currents, and — we shall argue in §1.4 (lecture 4) — the light states generally.

2.4 The symmetric orbifold: twisted sectors and long strings

Let $\mathcal{C}$ be a seed CFT of central charge c_1 (for $M_4 = T^4$: four bosons and four fermions per chirality, $c_1 = 6$). The Hilbert space of $\mathrm{Sym}^N(\mathcal{C})$ decomposes into twisted sectors labelled by conjugacy classes of S_N, i.e. by partitions of N,

$$[g] = (1)^{N_1}(2)^{N_2}\cdots(n)^{N_n}\cdots, \qquad \sum_n nN_n = N. \tag{54}$$

A cycle of length n identifies n copies cyclically: on such a cycle the fields obey $X_a(\sigma + 2\pi) = X_{a+1}(\sigma)$, so a single field is periodic only after n turns. The n copies have been glued into one "long string" of length n, with

$$\boxed{\text{mode spacing } \tfrac{1}{n}, \qquad \text{total central charge } n\,c_1, \qquad h_{\sigma_n} = \bar{h}_{\sigma_n} = \frac{c_1}{24}\Big(n - \frac{1}{n}\Big).} \tag{55}$$

The last is the dimension of the bare twist field σ_n [25, 26], obtained by summing the shifted zero-point energies of the fractional modes.

Three uses of (55).

The Cardy window. Repeating the estimate of §1.3: a state at level E_L of the full theory, realised in the maximally twisted sector $n = N$, has effective level NE_L in a CFT of central charge c_1, so the Cardy entropy of the long string is

$$S = 2\pi\sqrt{\frac{c_1\,(NE_L)}{6}} = 2\pi\sqrt{\frac{(Nc_1)E_L}{6}} = 2\pi\sqrt{\frac{c\,E_L}{6}}, \tag{56}$$

[checked] — the same formula — but now valid when $NE_L \gg c_1$, i.e. $E_L \gg c/N^2$, instead of $E_L \gg c$.

The threshold. Set $n = N$ in (55):

$$\Delta_{\sigma_N} = 2\cdot\frac{c_1}{24}\Big(N - \frac{1}{N}\Big) \;\longrightarrow\; \frac{c_1 N}{12} = \frac{c}{12} \qquad (N\to\infty), \tag{57}$$

[checked] i.e. $E_L = E_R \to 0$: the ground state of the maximally twisted sector sits exactly at the BTZ threshold $M = 0$. States above it lie in the black-hole range $E > 0$, states below it in the conical-defect range $-c/12 \le E < 0$. Excitations of the long string are therefore precisely the states that can be black holes, which is why it controls black hole thermodynamics.

The partition function. Because the twisted sectors are labelled by partitions, the partition functions of $\mathrm{Sym}^N(\mathcal{C})$ for all N are packaged in one generating function [23, 24]

$$\sum_{N\ge 0} p^N Z_N(q,\bar{q}) = \prod_{n>0}\prod_{h,\bar{h}}\Big(1 - p^n\, q^{(h-c_1/24)/n}\,\bar{q}^{(\bar{h}-c_1/24)/n}\Big)^{-d_1(h,\bar{h})\,\delta^{(n)}_{h-\bar{h}}}, \tag{58}$$

with d_1 the seed degeneracies and $\delta^{(n)}_{h-\bar{h}} = 1$ if $h - \bar{h} \equiv 0 \bmod n$ and zero otherwise (integer spin on the long string). The right-hand side is a product over single long strings of every length n carrying every seed excitation: a free gas of strings. This is what a second-quantised string theory looks like, and it is the technical basis of §2.3 (lecture 4).

Exercise 2. (i) Verify $h_{\sigma_2} = \bar{h}_{\sigma_2} = 1/4$ for $c_1 = 6$, and that $G_{-1/2}\bar{G}_{-1/2}\sigma_2$ has $(h,\bar{h}) = (1,1)$, as claimed in (53). (ii) Expand (58) to order p^2 for a seed with a single $(h,\bar{h}) = (0,0)$ state and check that you recover the untwisted S_2-invariant states together with the $\mathbb{Z}_2$-twisted sector.

2.5 The stringy exclusion principle

By (47) the D1–D5 theory carries an affine $\mathfrak{su}(2)$ R-symmetry at level $k' = N$. Unitarity of an affine $\mathfrak{su}(2)_{k'}$ representation bounds the spin of its *primary*,

$$j \;\le\; \frac{k'}{2} = \frac{N}{2}. \tag{59}$$

Chiral primaries of the small $\mathcal{N} = 4$ algebra are annihilated by all positive modes, hence are affine primaries, and they satisfy $h = j$; so (59) bounds their dimensions too. (Ordinary descendants, obtained by acting with J^+_{-1}, do reach larger J^3_0; the bound is on the primaries.) In supergravity, by contrast, BPS states are built by Kaluza–Klein reduction on S^3 and by taking multi-particle states of the result; the single-particle tower is unbounded and multi-particle states can be piled up freely. Supergravity taken literally therefore predicts chiral primaries of arbitrarily large j.

Maldacena and Strominger [15] showed that the two agree for particle numbers up to of order c, and that beyond that the CFT bound (59) has no counterpart in perturbative supergravity: in their words, it is "a stringy exclusion principle which cannot be seen in perturbation theory about AdS_3". It is a prediction *from* the CFT that the bulk must possess a non-perturbative mechanism cutting off the tower, not a check that the bulk already does. The mechanism was identified later: a graviton with large angular momentum on S^3 expands into a spherical brane — a giant graviton — whose size is bounded by the sphere, capping the tower at $j = N/2$ [16].

The pattern — the CFT knows c is finite, classical supergravity does not — recurs. It is the same finiteness that makes the sparseness condition of §1.3 (lecture 4) a non-trivial constraint.

2.6 Seiberg–Witten: where the CFT is singular

Not every point of (48) defines an acceptable CFT.

The instanton moduli space $\mathcal{M}_{Q_1,Q_5}$ has *small-instanton singularities*: loci where an instanton shrinks to zero size. In gauge theory language these are the points where the Higgs branch meets the Coulomb branch, i.e. where a D1-brane can leave the D5-branes and escape into the transverse $\mathbb{R}^4$. A sigma model whose target has such a singularity, with the escape direction costing no energy, develops an infinitely long throat with a linear dilaton. Seiberg and Witten [20] showed that the spectrum then contains a *continuum above a gap*, and computed the gap in the S-dual NS1/NS5 description (§3):

$$\Delta_0 = \frac{(Q_5 - 1)^2}{4Q_5}. \tag{60}$$

This is the standard linear-dilaton gap $\Delta_0 = Q^2/8$ with background charge $Q = \sqrt{2}(Q_5 - 1)/\sqrt{Q_5}$, read off from the improvement term in the stress tensor of the long-string theory [20, eq. (4.29)] [checked].

A continuum is fatal for everything we have been doing. Cardy's formula counts states; with a continuum there is nothing to count, the partition function does not converge as a discrete sum, and ρ is a spectral density rather than a degeneracy. This is the same disease as Liouville in lecture 2, §5.2.

Which points are singular. In the ADHM description of a single small instanton — a $U(1)$ gauge theory with Q_5 charge-one hypermultiplets (A^i, B_i) — the vacuum equations are

$$\sum_i A^i B_i = 0, \qquad \sum_i |A^i|^2 - \sum_j |B_j|^2 = \zeta, \tag{61}$$

with ζ a Fayet–Iliopoulos parameter. For $\zeta \neq 0$ the moduli space is the smooth $T^*\mathbb{CP}^{Q_5-1}$, whose zero section has radius $\sqrt{\zeta}$; at $\zeta = 0$ that $\mathbb{CP}^{Q_5-1}$ collapses and the space acquires a conical singularity [20, §5.1]. The FI parameter is a RR modulus of the D1–D5 system. The singular locus is therefore where the RR moduli vanish, which is precisely the *pure NS–NS* point; generic points of $\mathcal{M}$ are fine.

The missing chiral primaries. The concrete inconsistency Seiberg and Witten resolved is the following. (In this paragraph, and in [20], Δ denotes the *left-moving* weight $h = j$ of a chiral primary, not $h + \bar{h}$.) Worldsheet computations in the NS1/NS5 background [19], which by construction set all RR moduli to zero, find chiral operators only up to

$$\Delta \;\leq\; \frac{Q_5 - 2}{2}, \tag{62}$$

whereas the geometric computation — counting harmonic forms on $\mathcal{M}_{Q_1,Q_5}$, valid at generic moduli — gives chiral primaries up to $\Delta \leq Q_1 Q_5/2$, in agreement with (59) [17, 15]. Neither computation is wrong: they are done at different points of $\mathcal{M}$. The resolution [20, §5.1] is that the first "missing" chiral primary is the one associated with the L^2 harmonic form ω Poincaré dual to the zero section $\mathbb{CP}^{Q_5-1} \subset T^*\mathbb{CP}^{Q_5-1}$. That submanifold has complex codimension $Q_5 - 1$, so ω has bidegree $(Q_5 - 1, Q_5 - 1)$ and corresponds to a chiral primary of weight $\left(\frac{Q_5-1}{2}, \frac{Q_5-1}{2}\right)$ — exactly the quantum numbers of the first state absent from the worldsheet count. Its support sits at $|A|, |B| \sim \sqrt{\zeta}$; as $\zeta \to 0$ it runs down the linear-dilaton throat to $\phi \to \infty$ and ceases to be normalisable. The state has not disappeared: it has joined the continuum.

This is also the promised caveat on the protectedness of the index. An index is invariant under continuous deformations *provided* no states escape to infinity in field space, and at the singular locus that is exactly what happens.

Exercise 3. Check that (60) vanishes at $Q_5 = 1$ and equals $1/8$ at $Q_5 = 2$. Seiberg and Witten read the first fact as saying that at $Q_5 = 1$ the ground state itself becomes non-normalisable [20, §4]; the second reappears in §3.3. What happens to the "missing" chiral primary at $Q_5 = 1$?

3 Pure NS–NS flux: the S-dual corner

The D1–D5 description is useless for worldsheet computations: the background carries RR flux, for which no tractable worldsheet formulation exists. There is, however, a corner of the same conformal manifold (48) where the worldsheet theory is a WZW model. This section sets up that corner and records what is known about its boundary dual; we return to it in more detail later in the course.

3.1 The NS1–NS5 system and its near-horizon limit

Type IIB has an $\mathrm{SL}(2,\mathbb{Z})$ S-duality acting on the string coupling as $g_s \to 1/g_s$ and exchanging

$$\text{D1} \;\longleftrightarrow\; \text{F1 (fundamental string)}, \qquad \text{D5} \;\longleftrightarrow\; \text{NS5}. \tag{63}$$

Applied to the configuration of §1.1 it gives $p \equiv Q_1$ fundamental strings wound on S^1_y inside $k \equiv Q_5$ NS5-branes wrapping $S^1_y \times M_4$. The construction mirrors the D1–D5 one: harmonic functions on

the transverse $\mathbb{R}^4$, a black string in six dimensions, and a near-horizon $\mathrm{AdS}_3 \times S^3 \times M_4$. The string-frame solution is

$$\mathrm{d}s^2_{\text{string}} = H_1^{-1}\big(-\mathrm{d}t^2 + \mathrm{d}y^2\big) + H_5\big(\mathrm{d}r^2 + r^2\mathrm{d}\Omega_3^2\big) + \mathrm{d}s^2_{M_4}, \qquad e^{2\phi} = g_s^2\,\frac{H_5}{H_1}, \tag{64}$$

with

$$r_5^2 = k\,\alpha', \qquad r_1^2 = \frac{g_s^2\alpha'\,p}{v}. \tag{65}$$

The NS5 radius follows from flux quantisation and is independent of g_s; the F1 coefficient is the D1 one of (11) with an extra power of g_s, because the fundamental string couples to gravity with strength $T_{\mathrm{F1}}\kappa^2 \sim g_s^2\alpha'^3$ rather than $T_{\mathrm{D1}}\kappa^2 \sim g_s\alpha'^3$ [checked].

Three features distinguish this description from the D1–D5 one.

The dilaton. Compare $e^{2\phi} = g_s^2 H_5/H_1$ here with $e^{2\phi} = g_s^2 H_1/H_5$ for D1–D5. In the near-horizon limit $H_i \to r_i^2/r^2$ both become constant, here

$$e^{2\phi} \;\longrightarrow\; g_s^2\frac{r_5^2}{r_1^2} = \frac{kv}{p}, \qquad \Longrightarrow \qquad g_6^2\big|_{\text{near hor.}} = \frac{e^{2\phi}}{v} = \frac{k}{p} \tag{66}$$

[checked]. This matters: an NS5-brane throat *on its own* has a running dilaton — that is the little-string-theory throat, and it is the same throat that reappeared in §2.6 — and it is the fundamental strings that cap it off and produce AdS_3 with a constant dilaton. A constant dilaton means a genuine, tunable string coupling in the near-horizon region, so string perturbation theory makes sense there.

The radius. Reading off the near-horizon limit of (64),

$$\mathrm{d}s^2 \;\longrightarrow\; \frac{r^2}{r_1^2}\big(-\mathrm{d}t^2 + \mathrm{d}y^2\big) + \frac{r_5^2}{r^2}\mathrm{d}r^2 + r_5^2\,\mathrm{d}\Omega_3^2, \tag{67}$$

which is $\mathrm{AdS}_3 \times S^3$ with equal radii $\ell = r_5$, i.e.

$$\boxed{\frac{\ell^2}{\alpha'} = k} \tag{68}$$

[checked]. In the D1–D5 description the corresponding quantity was $\ell^2/\alpha' = g_6\sqrt{Q_1Q_5}$, a function of the continuous modulus g_6. There is no contradiction: S-duality rescales the string metric by $e^{-\phi}$, so the two ℓ^2's are radii in different string frames, and the two expressions in any case refer to different points of $\mathcal{M}$. The statement that matters is that on the NS–NS locus the radius in string units is an integer, fixed by flux quantisation and independent of the string coupling, which is what allows the worldsheet theory to be a WZW model at a fixed level.

Consistency of the central charge. Feeding (68) and (66) into the master formula (29),

$$c = \frac{6\ell^4}{g_6^2\alpha'^2} = \frac{6\,k^2\alpha'^2}{(k/p)\,\alpha'^2} = 6\,k\,p, \tag{69}$$

[checked], which is $6Q_1Q_5$: the same central charge as in (30), as it must be, since c is fixed by the small $\mathcal{N} = 4$ algebra and cannot change across $\mathcal{M}$.

The two bulk expansions are now controlled by two different integers,

$$\frac{\alpha'}{\ell^2} = \frac{1}{k} \quad (\alpha' \text{ corrections}), \qquad g_6^2 = \frac{k}{p} \quad (\text{genus expansion}), \qquad \frac{G_3}{\ell} \sim \frac{1}{c} = \frac{1}{6kp}. \tag{70}$$

At fixed k the $1/N$ expansion of the boundary theory is the genus expansion with $N = p$. Small k gives a very stringy bulk with a well-controlled genus expansion — the opposite of the usual supergravity limit, and the reason this corner is tractable.

3.2 The worldsheet theory and its spectrum

Because the flux is NS–NS, the sigma model on $\mathrm{AdS}_3 \times S^3 \times T^4$ is a WZW model,

$$\mathfrak{sl}(2,\mathbb{R})_k \ \oplus \ \mathfrak{su}(2)_k \ \oplus \ (T^4), \tag{71}$$

in the $\mathcal{N} = 1$ supersymmetric sense (the bosonic levels are $k+2$ and $k-2$). The spectrum was determined by Maldacena and Ooguri [31, 32, 33]. Three features matter here.

(a) **Spectral flow.** The physical Hilbert space is built from w-spectrally-flowed images of $\mathfrak{sl}(2,\mathbb{R})$ representations, $w \in \mathbb{Z}$ being the winding of the string around the boundary circle of AdS_3. The flowed representations are new representations, not rearrangements of the unflowed ones, and including them is required for modular invariance of the one-loop partition function [32].

(b) **Two kinds of representation.** Discrete series $\mathcal{D}^{\pm}_j$ with $\frac{1}{2} < j < \frac{k-1}{2}$ describe *short strings*: normalisable states localised in the interior of AdS_3, with a discrete spectrum of spacetime dimensions. Continuous series $\mathcal{C}^{\alpha}_j$ with $j = \frac{1}{2} + is$, $s \in \mathbb{R}$, describe *long strings*: configurations that can be pushed out towards the boundary at finite cost, with a continuous spectrum of spacetime dimensions above a gap.

(c) **The gap.** The long-string continuum in the spacetime spectrum starts at the Seiberg–Witten value (60), $\Delta_0 = (k-1)^2/4k$, and the upper bound $j < \frac{k-1}{2}$ on the discrete series marks where a short string ceases to be normalisable and becomes a long one.

The long strings are the same objects as the twisted sectors of §2.4: a string wound w times around the boundary circle is an operator of twist w. Giveon, Kutasov and Seiberg [30] identified the effective theory on a single long string: a linear-dilaton (Liouville-like) direction ϕ — the radial position of the string — together with $\mathfrak{su}(2)_{k-2}$ currents, four free fermions, and T^4, with

$$c_{\text{long string}} = \underbrace{6(k-1)}_{\phi,\ \mathfrak{su}(2)_{k-2},\ \text{fermions}} + \underbrace{6}_{T^4} = 6k, \tag{72}$$

[20, eq. (4.30)], and chiral operators $\mathcal{O}_j = e^{j\sqrt{2/k}\,\phi} V_j$ with $j \le (k-2)/2$ [20, eq. (4.31)] — the ones met in §2.6. Note that $c_{\text{long string}} = 6k$ is precisely the seed central charge that a symmetric orbifold with $N = p$ copies needs in order to reproduce (69).

3.3 The boundary dual

The observation of the previous paragraph — that the long-string theory has exactly the central charge of a symmetric-orbifold seed — has been made precise. The answer takes two forms depending on k, and it is convenient to treat them together because the difference between them is entirely the presence or absence of the Seiberg–Witten continuum.

$k = 1$: the tensionless string and the free symmetric orbifold

At $k = 1$ the RNS description (71) degenerates: the $\mathfrak{su}(2)$ level would be $k-2 = -1$. Consistently, the long-string theory (72) collapses to $c_{\text{seed}} = 6$ with no linear dilaton at all, and the gap (60) vanishes: a single NS5-brane produces no throat for a string to escape down [20, §4], so there is no continuum. Using instead the hybrid formalism, in which the $\text{AdS}_3 \times S^3$ factor is described by the $\mathfrak{psu}(1,1|2)_1$ WZW model, Eberhardt, Gaberdiel and Gopakumar [34, 35] showed

$$\boxed{\begin{array}{c}\text{IIB on } \text{AdS}_3 \times S^3 \times T^4, \\ \text{pure NS–NS, } k = 1, \ p \text{ strings}\end{array} \quad \Longleftrightarrow \quad \begin{array}{c}\text{the } \textit{free} \text{ symmetric orbifold} \\ \text{Sym}^N(T^4), \quad N = p \to \infty\end{array}} \tag{73}$$

The supporting evidence is unusually strong.

- *No continuum, and a state-by-state match.* The $\mathfrak{psu}(1,1|2)_1$ spectrum contains no continuous representations, and the full string spectrum matches the large-N symmetric orbifold state by state [35].
- *Correlators localise.* In [36] the worldsheet correlation functions are shown to localise on the moduli space of holomorphic covering maps from the worldsheet to the boundary sphere, branched over the operator insertions. That is precisely the combinatorial structure by which symmetric-orbifold correlators are computed [26]: the covering surface is the worldsheet of the long string. The worldsheet genus expansion becomes the $1/N$ expansion of the orbifold. This is the closest thing available to a derivation, rather than a check, of an AdS/CFT duality.

This is consistent with §2.3: at $k = 1$ one has $\ell^2 = \alpha'$, so the bulk string is tensionless at the AdS scale, and the boundary theory is free, with its infinite tower of higher-spin currents (51). The two statements are the same statement.

$k \geq 2$: a deformed symmetric orbifold

For $k \geq 2$ the continuum is present, so the dual cannot be a symmetric orbifold of a compact seed. Eberhardt [38] proposed, and checked to fourth order in conformal perturbation theory at large N (the spectra of short and long strings, and two- and three-point functions against the genus-zero worldsheet result), that

$$\boxed{\begin{array}{c}\text{IIB on } \text{AdS}_3 \times S^3 \times T^4, \\ \text{pure NS–NS, } k \geq 2\end{array} \quad \Longleftrightarrow \quad \begin{array}{c}\text{Sym}^N\big(\mathbb{R}_Q \times T^4\big), \quad c_{\text{seed}} = 6k, \\ \text{deformed by a twist-two marginal operator}\end{array}} \tag{74}$$

where $\mathbb{R}_Q$ denotes an $\mathcal{N} = 4$ linear-dilaton theory whose background charge is fixed by k [37, 38]. Three points:

- the seed is exactly the long-string theory (72) of Giveon–Kutasov–Seiberg, with $c_{\text{seed}} = 6k$, so that $c = 6kp$ as required;
- it is *non-compact*, which is how the Seiberg–Witten continuum survives — the linear-dilaton direction is the throat of §2.6, and its gap is (60);
- the deformation is again of twist two, as in (53).

So the symmetric-orbifold structure with a twist-two deformation is not a peculiarity of the free point: it persists along the whole NS–NS locus, with the seed varying with k and degenerating to the compact T^4 at $k = 1$.

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